Boyuan Zhang

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Education

Stanford University — *Sep. 2024 – June 2026 (exp.)*
M.S., Statistics

- **Coursework** (all PhD–level): Causal Inference, Bayesian Statistics, Convex Optimization, Econometrics, Theoretical Statistics, Applied Statistics, Probability Theory, Stochastic Methods, Matching Theory, Numerical Linear Algebra, Social Data Analysis.

New York University — *Jan. 2021 – Dec. 2023*
B.A. (cum laude), Computer and Data Science

- Honors: University Honors Scholar, Founders' Day Award, Dean's List (every academic year).
- **Coursework** (* graduate level): Mathematical Statistics*, Applied Statistics*, Probability and Statistics*, Stochastic Processes*, Statistical Learning Theory*, ML for Healthcare*, Advanced ML and Deep Learning, Causal Inference, Data Management and Analysis, Responsible Data Science, Algorithms, Real Analysis.

Research Interests

- **Methods:** Causal Discovery and Inference, Statistical Inference, Machine Learning, Experimental Design;
- **Applications:** Healthcare, Public Policy, Social Science, and Sports.

Preprints

Treatment Response as a Latent Variable — In Submission at JRSSB
Christopher Tosh, ***Boyuan Zhang***, Wesley Tansey
arXiv:2502.08776

Automated Discovery of Difference-in-Differences — Under Revision
William Herlands, ***Boyuan Zhang***, Edward McFowland III, Daniel Neill

Research Experience

Stanford University, Department of Statistics — *Stanford, CA*
Independent Study — *June 2025 – Present*

- **Research Area:** Causal Mediation Analysis, Causal Representation Learning, Generative Models
- **Advisor:** Dr. Wing Hung Wong
- Proposed a joint learning framework for causal mediation analysis, enabling flexible dimension reduction and causal effect estimation under high-dimensional confounding using generative models.
- Learned latent representations capturing dependencies among treatments, mediators, and outcomes, facilitating identification of direct and indirect effects across binary, continuous, and multivariate settings.

Stanford Law School, Deborah L. Rhode Center on the Legal Profession — *Stanford, CA*
Research Assistant — *June 2025 – Present*

- **Research Area:** Policy Learning, Mobility Analysis, Hierarchical Clustering, Eviction Outcomes
- **Advisor:** Dr. Aviv Caspi
- Developed algorithms to detect household mobility patterns from device-level location data, evaluating pre- and post-eviction dynamics in Los Angeles County.
- Linked household mobility outcomes with eviction events to evaluate displacement effects and inform evidence-based housing and legal policy interventions.

Stanford University, Department of Management Science and Engineering — *Stanford, CA*
Research Assistant — *March 2025 – Present*

- **Research Area:** Kidney Allocation, Policy Learning, Market Design, Strategic Decision Making
- **Advisor:** Dr. Itai Ashlagi
- Conducted empirical analyses of kidney allocation data to evaluate efficiency–equity trade-offs, enhance transparency, and inform policy design in organ allocation.
- Developed computational models to assess allocation policies and examine dynamic incentives of transplant centers' acceptance preferences, expedited organ offer mechanisms, and patients' strategic listing behaviors.

Memorial Sloan Kettering Cancer Center — *New York, NY*
Research Intern, Department of Epidemiology & Biostatistics — *June 2023 – May 2024*

- **Research Area:** Multiple Testing, Empirical Bayes, Non-compliance, False Discovery Rate Control
- **Advisor:** Dr. Wesley Tansey
- Developed a nonparametric testing procedure for individual causal effects from observational data with potential confounding and non-compliance.
- Demonstrated effective control of the false discovery rate at the target level in simulations while maintaining reasonable power compared to BART, Causal Forest, and FDR Regression.
- Discovered biologically rational markers that predispose a cell line to be sensitive or resistant to treatment with a targeted agent using data from a large cancer drug study.

New York University, Courant Institute of Mathematical Science — *New York, NY*
Research Assistant — *Feb. 2023 – May 2024*

- **Research Area:** Difference-in-Difference, Subset Scanning, Change Point Detection, Subgroup Selection
- **Advisor:** Dr. Daniel Neill
- Refined and improved the subset scanning algorithms for automated difference-in-difference discovery.
- Identified both heterogeneous treatment discontinuities in high-dimensional categorical data and appropriate control subsets with statistical and econometric validity.
- Demonstrated robust performance in simulations, and discovered novel policy-relevant insights in complex public policy datasets.

Carnegie Mellon University, Department of Statistics and Data Science — *Pittsburgh, PA*
Summer Undergraduate Research Experience in Statistics — *June 2022 – July 2022*

- **Research Areas:** Regularized Regression, Bayesian Hierarchical Modeling, Sports Analytics
- **Advisors:** Dr. Ronald Yurko, Dr. Konstantinos Pelechrinis
- Developed a regularized regression model for evaluating player's performance and contribution in soccer.
- Incorporated priors learned from box score statistics into a regularized adjusted plus-minus model framework.
- Demonstrated better predictability and interpretability than existing methods through simulations.

Teaching Experience

New York University, Center for Data Science & Courant Institute — *New York, NY*
Teaching Assistant — *Sep. 2022 – Dec. 2023*

DS-UA 301 Advanced Techniques in Machine Learning and Deep Learning — Fall '23
DS-UA 201 Causal Inference — Summer '23
CSCI-UA 2 Intro to Computer Programming — Fall '22 '23, Spring '23

Selected Course Projects

Analyzing the Impact of Subway Major Incidents on Passenger Flows — Summer 2023
Modeling passenger dynamics within subway network using turnstile data

Collaborative Filtering (CF) Algorithms for Recommendation Systems — Spring 2023
Comparative Analysis of Matrix Factorization, Neighborhood Models, and Neural CF

Calibrated NBA Win Probability Estimation — Spring 2023
Multi-objective RNN to estimate win probability for both accuracy and calibration

Fingertips Position Estimation of a Robot Hand — Fall 2022
CNN estimates the positions of the robot's fingertips through RGBD images

Lip Reading Word Classification — Fall 2022
CNN-LSTM-based classifier model identifies the speaker's word through video

Skills

Programming Languages: Python, R, SQL, Java, C/C++, Bash, MATLAB
Software/Tool: LaTeX, Git, Markdown, Tableau, MongoDB, Android Studio, Arduino, OCAD, Excel